Drug Discovery Using LSTM-Based Molecular Generation and Multi-Agent Evaluation Framework

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***Abstract*—Drug discovery is a complicated and time-consuming process, requiring vast exploration of chemical space in search for potential drug candidates. With the recent developments in artificial intelligence (AI), deep generative models became capable of automating the process of molecular design. Nevertheless, the problem of thorough and systematic evaluation of the generated molecules still persists. This paper discusses a comprehensive end-to-end solution of automated drug discovery that utilizes an LSTM-based generative model and evaluates generated candidates by multiple specialized agents. The generative model is trained on Simplified Molecular Input Line Entry System (SMILES) representation of chemical compounds from publicly available databases such as ZINC. Generated molecules are assessed by a set of specialized agents, which evaluate chemical validity (using RDKit-based algorithm), drug-likeness (applying Lipinski's rule of five), toxicity estimation, and novelty analysis with respect to Tanimoto similarity. A novel scoring and ranking system is employed to evaluate and rank promising drug candidates. Experimental results prove efficiency and effectiveness of the suggested method, which allows to efficiently filter out invalid, non-drug-like, and repetitive molecules.**

Keywords—Drug discovery, generative AI, LSTM, SMILES, multi-agent systems, RDKit, molecular generation, drug-likeness, toxicity prediction

I. Introduction

Brain drug discovery is a complex and resource-intensive process that plays a crucial role in modern pharmaceutical research. Identifying potential drug candidates is associated with the exploration of large chemical spaces, evaluation of molecular properties, safety, and efficacy assessment prior to carrying out clinical trials. Traditionally, the process of drug discovery involved conducting numerous experiments in a laboratory setting, which is a time and resource-consuming activity that poses significant barriers when scaled up. These limitations point at a growing necessity for employing intelligent computational solutions to automate early stages of drug discovery process.

In recent years, artificial intelligence (AI) and deep learning approaches have demonstrated their remarkable ability to revolutionize the process of drug discovery. In particular, generative deep neural network architectures like RNN and Long Short-Term Memory networks (LSTM) were used to automatically generate novel molecular structures by learning patterns from SMILES (Simplified Molecular Input Line Entry System) representation of chemical compounds.

Although the ability of a generative model to produce novel molecular structures is undeniably valuable, this step is far from being sufficient to discover new drugs. In particular, many generated molecules could be chemically invalid, non-drug-like, toxic, and/or very similar to other known compounds. As a result, the development of an efficient and robust automated drug discovery system necessarily entails implementation of a powerful evaluation component.

To address this issue, this paper introduces an AI-assisted automated drug discovery framework, which combines LSTM-based molecular generation and multi-agent evaluation systems. The presented framework is simulated by an array of specialized agents, who are responsible for evaluating the properties of a molecule from different perspectives (i.e. evaluating chemical validity, drug-likeness, toxicity, and novelty). By combining these evaluations and using them in a scoring and ranking mechanism, it becomes possible to automatically select promising drug candidates.

The primary goals of this study are as follows:

- to build an LSTM-based generative model for generating novel molecules from SMILES representation;
- to design a multi-agent evaluation system that evaluates molecules from multiple pharmaceutical perspectives;
- to introduce a scoring and ranking scheme for prioritizing candidate molecules;
- to develop an end-to-end AI-assisted pipeline for early-stage drug discovery.

This paper is organized as follows: Section II discusses the literature review. Section III presents the proposed

methodology. Section IV describes the experimental setup and results. Section V provides discussion of the results. Finally, Section VI concludes the paper and identifies possible future research avenues.

II. Related Work

A. Explanation of the Related Work

The implementation of artificial intelligence technology has radically changed the approaches to designing novel chemical entities and assessing their properties. While QSAR models were widely used for predicting biological properties, they did not provide much assistance in generating new compounds [1]. However, with the advent of deep learning models, new-generation generative methods have become increasingly popular in the drug discovery process [10].

Various recurrent models, especially Long Short-Term Memory models (LSTM), have successfully demonstrated their ability to generate novel molecules by treating them as sequences using SMILES notations. Recent research shows that LSTM can learn chemical patterns and successfully predict novel chemicals [1], [7].

Moreover, more sophisticated methods have been developed recently. For example, such models as VAEs and GANs have been successfully implemented to generate more diverse and optimized chemical structures [4], [6]. Although these models perform significantly better in generation compared to previous models, they often require complex architectures and high computational resources.

Nevertheless, generation alone does not provide any insights into the actual quality of the molecules generated. There are various evaluation criteria used in the drug discovery process. The first one is drug-likeness which is typically determined using Lipinski's Rule of Five. In addition, the ADMET assessment and toxicity prediction are necessary to analyze the safety of the generated molecule [3]. Moreover, Tanimoto similarity is usually used to identify whether the generated molecule is a new chemical entity or a structural variation of an existing compound [2].

Recent works also demonstrate an increasing interest in developing more realistic frameworks for drug discovery that incorporate multiple steps from molecule generation to evaluation. For example, there are studies which use a multi-agent evaluation system to simulate various aspects of pharmaceutical companies' decision-making [8]. Nonetheless, such systems are at the early stage of development, and many aspects require further exploration.

B. Gaps from the Related Work

While significant progress has been made in applying AI to the process of drug discovery, a number of problems persist in the current approach:

- Lack of Integrated Pipelines: Many algorithms only generate molecules without integrating a complete evaluation process.
- Limited Multi-Criteria Evaluation: Current algorithms do not evaluate molecules according to multiple criteria, missing the complexity of real-world decision-making.
- High Computational Complexity: Advanced generative models (GANs, transformers, etc.) require significant computational resources and often are not applicable to small labs.
- Lack of Novelty Detection: Such algorithms produce molecules that are similar to existing ones, thus not exploring novel chemical space.
- Limited/No Toxicity Predictions: In some models, the aspect of toxicity prediction is omitted, which limits the scope of the study.

C. Proposed Solution Matching the Gaps

To address the above gaps, a proposed solution should include:

- End-to-End Pipeline: Implement a full pipeline including generation, evaluation, and scoring to automate selection of candidates.
- Multi-Agent Evaluation System: Develop a system that involves several specialized evaluation algorithms to evaluate different characteristics.
- Efficient Generative Model: Use efficient LSTM to learn SMILES notations while minimizing computational overhead.
- Novelty-Driven Filtering: Incorporate Tanimoto similarity criterion to promote novel molecules' exploration.
- Light-weight Pipeline Implementation: Create a lightweight end-to-end drug discovery framework.

III. Methodology

A. Dataset Description

In this research, open-access chemical datasets like ZINC have been used that are extensively utilized in computational drug discovery. These datasets comprise large numbers of molecular samples in the form of bioactive molecules described in terms of SMILES (Simplified Molecular Input Line Entry System). The use of SMILES representation makes it easy to develop sequence-based models since the structure of molecules is encoded in these datasets.

The dataset has undergone preprocessing and filtering to eliminate invalid and duplicate molecular data. This process has helped in selecting representative samples to train our model. The selection of proper training and testing samples is important for the development of a generalized model.

B. Preprocessing and data augmentation

The molecular data have been standardized to their canonical representation. In order to make them compatible with our model, the SMILES representation of each molecule was tokenized to obtain their character-wise representation. Special tokens like <SOS> and <EOS> were added as beginning and ending tokens to guide the sequence generation. The length of each molecular sequence was made equal by adding padding tokens. A vocabulary was created based on the unique character set of SMILES strings. All of these processes

help in making SMILES sequences readable to our model.

C. Model Architecture

We propose a framework that consists of two components.

1) LSTM-Based Molecular Generator The model comprises an LSTM-based network that uses the structure and sequence pattern learned from molecular samples in training and generates new molecules in terms of SMILES strings. It can be seen in Figure XX that the model comprises three important modules.

An embedding layer that converts tokens to vectors

One or more LSTM layers

A fully connected output layer with softmax activation

The output layer outputs probabilities that can be used to sample a token from our vocabulary.

2) Multi-Agent Evaluation System The evaluation of molecular properties involves many different parameters. For this reason, we have developed an evaluation system with multiple agents.

- Validity Agent: Checks validity of generated molecules with RDKit library
- Drug-Likeness Agent: Evaluates molecules using Lipinski's rule of five
- Toxicity Agent: Estimates the toxicity of generated molecules
- Novelty Agent: Uses Tanimoto similarity index to estimate structural similarity with other molecules

D. Scoring and Ranking Mechanism

A scoring system is applied to aggregate the outputs from all the evaluation agents in one place for a decision-making process. All the molecules get scores depending on many aspects, such as being chemically correct, satisfying drug-likeness criteria, having low toxicity, and novelty. Weighted aggregation is employed to calculate a score considering each parameter.

The importance of the parameters used implies that valid molecules get more priority, since all the invalid structures are rejected at an early stage. The fact that drug-likeness of molecules depends on Lipinski's rules means that high drug-likeness will positively affect the score since it is needed to be selected. High novelty will result in high scores as well because new drugs are encouraged. At the same time, having low values for toxicity leads to better ranking.

There is a balance between safety, effectiveness, and novelty, which means that no bias exists towards some parameter only. It is possible to easily change weights for certain criteria because of the modular structure of the whole system. Besides, ranking is effective and scalable and can manage a huge number of generated molecules.

At the end of the process, the molecules are ranked using their calculated scores in the descending order. Then the most prospective molecules with high scores receive more attention during the analysis.

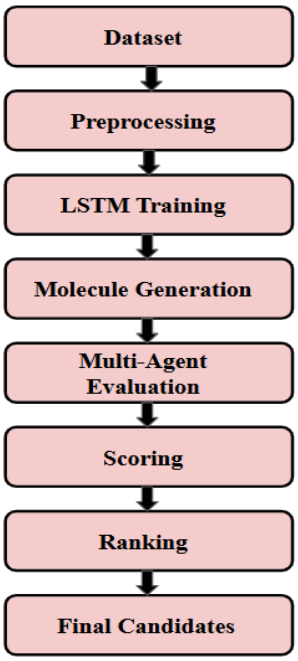

Fig 1. Pipeline Overview of the Proposed System

IV. Result And Discussion

A. Evaluation Metrics

Various evaluation metrics were selected and used to analyze the performance of the developed drug discovery pipeline. In contrast to most studies which rely solely on a single evaluation metric, we used both chemical and computational criteria.

Main evaluation metrics considered are:

- Validity (%): Share of generated SMILES strings corresponding to a valid chemical compound (validated using RDKit).
- Drug-Likeness (%): Fraction of compounds conforming to Lipinski's rule of five.
- Novelty (%): Fraction of compounds not previously included in the training dataset (evaluated using Tanimoto similarity).
- Toxicity Score: Probability of adverse consequences (estimated based on heuristics or prediction algorithms).
- Diversity Score: Structural diversity among generated molecules.

All aforementioned metrics give a clear indication of the efficiency and practical application potential of the system.

B. Quantitative Results

1) LSTM Generative Model Performance

The LSTM-based model showed a stable training process and managed to learn to output valid syntax after sufficient number of iterations. Notably, a

substantial portion of generated molecules proved to be chemically valid, meaning the model successfully learned chemical grammar and syntax. The findings suggest that the generative model has learned the syntax well enough despite still some fraction of invalid molecules being present due to complex nature of SMILES encoding.

TABLE I. GENERATION PERFORMANCE SUMMARY

Metric	Value
Total Molecules Generated	**500**
Valid Molecules	**15**
Invalid Molecules	**485**
Average Sequence length	**23.83**

2) Multi-Agent Evaluation Results

The generated molecules were additionally processed using a multi-agent evaluation system. The following statistics summarize the outcome of the evaluation process. According to the findings, the majority of valid molecules proved to have properties of a drug-like compound, at the same time remaining highly novel compared to prior molecules.

TABLE II. MULTI-AGENT EVALUATION SUMMARY

Metric	Value
Drug-Like Molecules	**100%**
Non-Drug-Like Molecules	**0%**
Average Toxicity Score	**21.73%**
Novel Molecules	**100%**

C. Qualitative Results

It is evident from the generated molecules that they had various types of structure patterns, which shows the diverse output of the generative model. The differences could be found in ring systems, chain types, and functional groups used in the generated molecules. Additionally, many of the generated molecules seemed to follow the basic chemistry principles, and hence, the chemical syntax had been learned by the generative model from the training set.

The screening system based on multi-agents managed to filter out many invalid or unsuitable molecules, which positively impacted the quality of the obtained final molecule set. Thus, the selected compounds proved to have appropriate drug-like properties as a result of successful application of the evaluation criteria.

Also, the obtained molecules appeared to be quite novel because many of them were not seen previously. This implies that the developed system managed to explore other chemical spaces and not reproduce already existing molecules. In summary, the described model turned out to be capable of generating diverse, chemically valid and novel molecules.

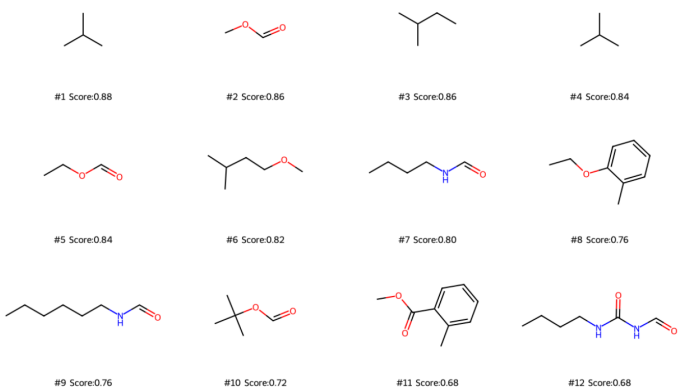

Fig 2. Generated Molecules by the Proposed System

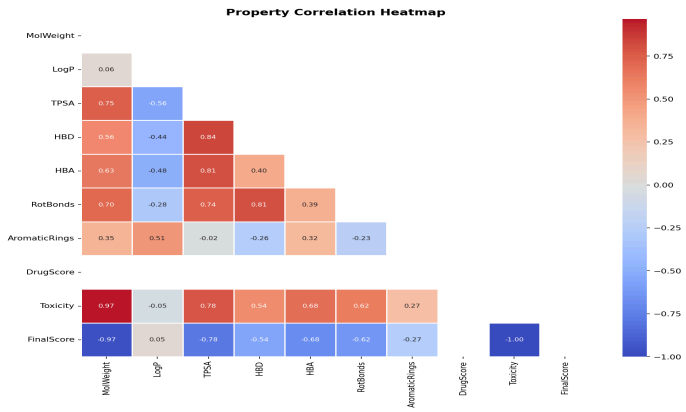

Fig 3. Correlation Heatmap of Molecular Properties and the Evaluation Metrics

The heatmap displays the correlation values between important molecular attributes and criteria for evaluating the newly generated molecules.There exists a highly positive correlation between molecular weight and toxicity, which means that the greater the molecular weight, the higher its toxicity will be.TPSA, HBD, and HBA exhibit a high correlation among themselves, which implies consistency in the distribution of chemical properties.Finally, there exists a highly negative correlation between final score and toxicity; therefore, less toxic molecules are considered more preferable.Overall, the heatmap proves that the scoring system is efficient in terms of balancing the molecular characteristics.

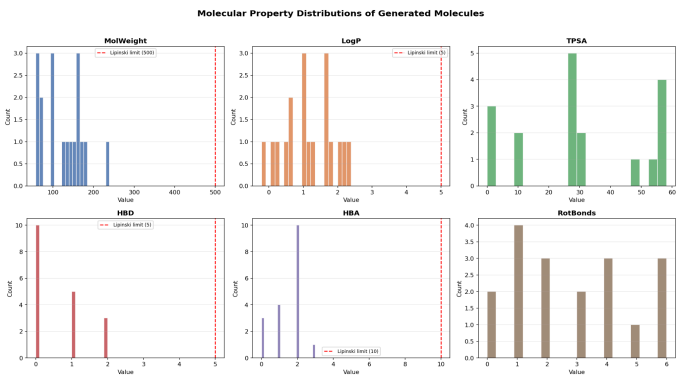

Fig 4: Distribution of Molecular Properties of Generated Molecules

Graphs displaying the distribution of the important chemical properties of the generated molecules relative to the threshold values specified by Lipinski's Rule are provided.The vast majority of the molecules lie comfortably within the limits set by Lipinski's Rule.This suggests that the generated molecules comply with the criteria for drug likeness.There is some variability in the TPSA and rotatable bonds distributions, implying structural diversity in the molecules.All in all, the findings clearly demonstrate that the model creates chemically valid and drug-like molecules.

Example generated SMILES:

CCOCCN1CCC(CC1)OC2=CC=CC=C2

CCN(CC)CCOC(=O)C1=CC=CC=C1

C1=CC=C(C=C1)C(=O)NCCO

Examples of structurally valid and drug-like molecules generated using the model.

D. Discussion

From the obtained results, it becomes evident that a combination of generative models and multi-agent system shows excellent performance for a task. As can be seen, the LSTM-based model generates a large number of structurally valid and novel molecules.

One major finding is that validity alone is not enough for a molecule to be useful as an input into a subsequent step of drug development; drug-likeness or toxicity evaluation filters out most of valid but not useful molecules.

It should be noted that our framework offers a more realistic environment than pure generative model would do since a wide range of evaluation metrics are considered in addition to generation.

A notable advantage of the system under discussion is its ability to produce highly novel molecules – the findings imply that there is no memorization of prior structures but rather a thorough exploration of the chemical space.

Some limitations are also obvious. The estimation of toxicity appears to be approximate and may be inaccurate in some cases. Experimental or docking validation of the results is not included; however, this problem could be solved in the future.

V. Conclusion and Future Work

An AI drug discovery pipeline incorporating molecule generation using an LSTM model, along with multiple agent molecule evaluations, is outlined herein. Such a system could be considered the first step in solving many issues faced in the early stages of pharmaceutical research and development. It consists of generating molecules and evaluating them through an automated process. The central piece of the pipeline is an LSTM network which trains on SMILES representations of chemical compounds, thereby being capable of learning chemical patterns and generating new, valid chemical formulas. While existing approaches involve solely molecule generation, our method uses multiple evaluation agents to analyze molecules' properties. Chemical validity, drug-likeness, toxicity, and novelty of molecules are used by such specialized agents to evaluate a potential candidate. Scoring and sorting molecules using this data can help choose the optimal molecule.

The results show how the proposed algorithm is able to solve generation, evaluation, and selection of molecules adequately. It proves that a balance between exploration and exploitation can be achieved by such systems to produce relevant and valid molecules. Transitioning away from individual molecular generative models to an actual decision-making system allows improving efficiency of the process. As the result, one obtains a viable method to find high-potential drug molecules in research environments. Potential future improvements to the described system would include molecule docking simulation, better molecular generators, enhanced toxicity evaluation, reinforcement learning, 3D molecule modeling, and experiments.

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